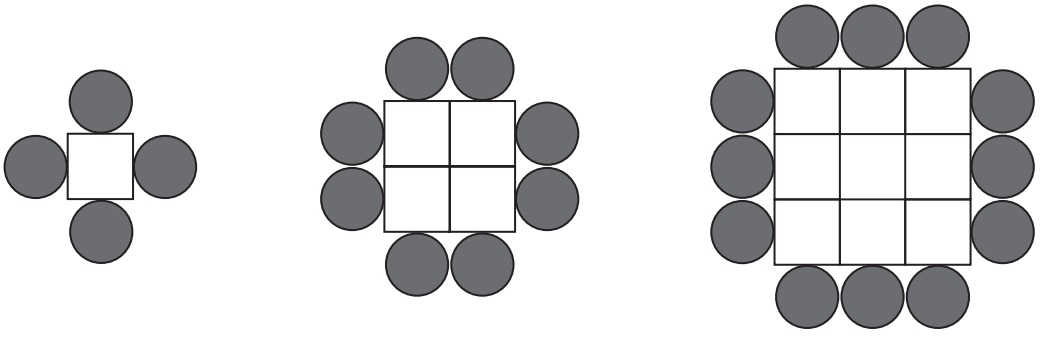
11 A sequence of patterns is made from circular tiles and square tiles

Here are the first three patterns in the sequence.



pattern number 1 pattern number 2 pattern number 3

(a) How many square tiles are needed to make pattern number 6?

.....

(2)

(b) How many circular tiles are needed to make pattern number 20?

.....

(2)

Derek says,

"When the pattern number is odd, an odd number of square tiles is needed to make the pattern."

(c) Is Derek right?
You must give reasons for your answer.

.....

.....

(2)

(Total for Question 11 is 6 marks)